HEALTHCARE RE-ADMISSON ANALYSIS

SQL QUERIES:

-- Checking weather the database is connected to the sql server

1. select * FROM deep_cleaned_patient_data limit 15;

-- Readmission overview

-- What is the overall readmission distribution?

-- How many records are NO, <30, and >30?

2. SELECT readmitted, COUNT(*) AS cnt

FROM deep_cleaned_patient_data

GROUP BY readmitted;

-- Demographic impact Which gender has the highest readmission rate?

-- Which race group has the highest readmission rate?

-- Which age group is most likely to be readmitted?

3. SELECT gender, readmitted, COUNT(*) AS cnt

FROM deep_cleaned_patient_data

GROUP BY gender, readmitted;

-- Hospital stay and utilization Does longer time_in_hospital correlate with higher readmission?

-- Do patients with more number_inpatient visits get readmitted more often?

-- How does number_emergency relate to readmission?

4. SELECT time_in_hospital, AVG(CASE WHEN readmitted <> 'NO' THEN 1 ELSE 0 END) AS readmit_rate

FROM deep_cleaned_patient_data

GROUP BY time_in_hospital

ORDER BY time_in_hospital;

-- Diagnosis patterns What are the top primary diagnoses (diag_1) for readmitted patients?

-- Which secondary diagnoses (diag_2, diag_3) appear most often in readmitted cases?

-- Which diagnosis codes have the highest proportion of readmission?

5. SELECT diag_1, readmitted, COUNT(*) AS cnt

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FROM deep_cleaned_patient_data
GROUP BY diag_1, readmitted
ORDER BY diag_1, cnt DESC;
-- Medical specialty
-- Which medical_specialty has the highest readmission count?
-- Which specialties have the highest percentage of <30 readmissions?
6. SELECT medical_specialty,
       SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) AS readmit_under_30,
       COUNT(*) AS total
FROM deep_cleaned_patient_data
GROUP BY medical_specialty
ORDER BY readmit_under_30 DESC;
-- Admission and discharge
-- Which admission_type_id is most common for readmitted patients?
-- Which discharge_disposition_id has the highest readmission rate?
-- Which admission_source_id sends the most readmitted patients?
7.SELECT admission_type_id, readmitted, COUNT(*) AS cnt
FROM deep_cleaned_patient_data
GROUP BY admission_type_id, readmitted;
-- Medication and treatment
-- Does diabetesMed = 'Yes' affect readmission?
-- How does insulin usage relate to readmission?
-- Are medication changes (change) associated with more readmissions?
8. SELECT 'change',
       SUM(CASE WHEN readmitted <> 'NO' THEN 1 ELSE 0 END) AS readmit_count,
       COUNT(*) AS total
FROM deep_cleaned_patient_data
GROUP BY 'change';
```

-- Lab results

-- How does A1Cresult distribution compare between readmitted and non-readmitted patients?

-- How does max_glu_serum relate to readmission?

9. SELECT A1Cresult, readmitted, COUNT(*) AS cnt

FROM deep_cleaned_patient_data

GROUP BY A1Cresult, readmitted;

-- Medication combinations

-- Which insulin or combination therapies appear most often in readmitted patients?

-- Compare readmission for patients on metformin, glipizide, or glimepiride.

10. SELECT insulin, COUNT(*) AS cnt

FROM deep_cleaned_patient_data

WHERE readmitted <> 'NO'

GROUP BY insulin

ORDER BY cnt DESC;

-- High-risk patient signals

-- Which age_mid groups have the worst readmission outcomes?

-- Which patients have high num_medications and also get readmitted?

11. SELECT age_mid,

AVG(num_medications) AS avg_medications,

AVG(CASE WHEN readmitted <> 'NO' THEN 1 ELSE 0 END) AS readmit_rate

FROM deep_cleaned_patient_data

GROUP BY age_mid

ORDER BY age_mid;

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-- These are especially useful for real healthcare analytics:
-- 1.Which age groups and genders have the highest <30 readmission rate?
-- 2.What are the top 5 primary diagnoses for patients readmitted in less than 30 days?
-- 3.Which medical specialties have the largest share of readmitted patients?
-- 4. How does time_in_hospital differ between readmitted and non-readmitted patients?
-- 5. Does diabetesMed = 'Yes' reduce or increase readmission?
-- 6. Are patients with change = 'Ch' more likely to be readmitted?
-- 7. Which admission source is associated with the most readmissions?
-- ANSWERS FOR EACH ONE
-- 1
SELECT
  age_mid,
  gender,
  SUM(CASE WHEN readmitted = '<30' THEN 1 ELSE 0 END) AS under30_count,
  COUNT(*) AS total,
  (SUM(CASE WHEN readmitted = '<30' THEN 1.0 ELSE 0 END) / COUNT(*)) * 100 AS
under30_rate_pct
FROM deep_cleaned_patient_data
GROUP BY age_mid, gender
ORDER BY under30_rate_pct DESC, under30_count DESC;
-- 2
SELECT
  diag_1 AS primary_diagnosis,
  COUNT(*) AS cnt
FROM deep_cleaned_patient_data
WHERE readmitted = '<30'
GROUP BY diag_1
ORDER BY cnt DESC
LIMIT 5;
```

```sql
-- 3
WITH total_read AS (
  SELECT COUNT(*) AS total_readmitted
  FROM deep_cleaned_patient_data
  WHERE readmitted <> 'NO'
)
SELECT
  medical_specialty,
  COUNT(*) AS readmitted_count,
  ROUND((COUNT(*) * 100.0) / tr.total_readmitted, 2) AS pct_of_all_readmissions
FROM deep_cleaned_patient_data, total_read tr
WHERE readmitted <> 'NO'
GROUP BY medical_specialty, tr.total_readmitted
ORDER BY readmitted_count DESC;
-- 4
SELECT
  CASE WHEN readmitted = 'NO' THEN 'Not Readmitted' ELSE 'Readmitted' END AS
readmitted_flag,
  COUNT(*) AS n,
  ROUND(AVG(time_in_hospital),2) AS avg_days,
  ROUND(STDDEV_POP(time_in_hospital),2) AS sd_days,
  MIN(time_in_hospital) AS min_days,
  MAX(time_in_hospital) AS max_days
FROM deep_cleaned_patient_data
GROUP BY readmitted_flag;
```

```sql
-- 5
SELECT
  diabetesMed,
  COUNT(*) AS total,
  SUM(CASE WHEN readmitted <> 'NO' THEN 1 ELSE 0 END) AS readmitted_count,
  ROUND((SUM(CASE WHEN readmitted <> 'NO' THEN 1.0 ELSE 0 END) / COUNT(*)) * 100,2) AS readmission_rate_pct
FROM deep_cleaned_patient_data
GROUP BY diabetesMed
ORDER BY readmission_rate_pct DESC;
-- 6
SELECT
  'change',
  COUNT(*) AS total,
  SUM(CASE WHEN readmitted <> 'NO' THEN 1 ELSE 0 END) AS readmitted_count,
  ROUND((SUM(CASE WHEN readmitted <> 'NO' THEN 1.0 ELSE 0 END) / COUNT(*)) * 100,2) AS readmission_rate_pct
FROM deep_cleaned_patient_data
GROUP BY 'change'
ORDER BY readmission_rate_pct DESC;
-- 7
SELECT
  admission_source_id,
  COUNT(*) AS readmitted_count
FROM deep_cleaned_patient_data
WHERE readmitted <> 'NO'
GROUP BY admission_source_id
ORDER BY readmitted_count DESC
LIMIT 10
```